

TryHackMe — Computer Fundamentals Module

Windows CLI Basics

Learning Notes — CMD Navigation, File Search, System Info & Network Commands

"Windows is one of the most widely used operating systems in workplaces worldwide. You will often investigate or troubleshoot Windows machines — especially user desktops and laptops." — TryHackMe, Room Introduction

 Navigation File Search System Info Network Cmds Process Mgmt Lab Mission

1. Introduction — The Windows Command Prompt

The Windows Command Prompt (CMD) is a text-based interface for interacting with the Windows operating system. Instead of clicking folders and menus, you type commands that tell the system exactly what you want to do — list files, move between folders, check system information, or query the network. This room is the Windows counterpart to Linux CLI Basics, using the same story-driven hands-on approach.

Day 2 Scenario

You arrive for your second day at your internship. Your supervisor greets you with a quick update and then leaves. You are on your own, faced with a terminal prompt and a mystery: find the `task_brief.txt` file somewhere in the user folder — using only the command line. Each step will guide you through exactly what cyber security professionals do every day.

1.1 CMD vs PowerShell vs Linux Terminal

Feature	CMD (Command Prompt)	PowerShell	Linux Terminal (bash)
Introduced	Windows 95 / NT era	Windows XP SP2 / Vista	Unix/Linux — 1970s onwards
Language	Batch scripting (cmd)	PowerShell scripting language	Shell scripting (bash/sh)

Capabilities	Basic navigation, tools	Full .NET integration, objects	Extensive scripting, pipes
Case sensitive	No — case insensitive	No (by default)	Yes — case sensitive
Objects/Text	Text output only	Object-based output	Text stream output
Security work	Basic recon, file search	Advanced automation, WMI, AD	Primary tool for Linux servers
Availability	All Windows versions	Windows 7+ (pre-installed W10+)	Not native on Windows

How to Open Command Prompt

Win + R → type cmd → Enter | Search bar → type cmd → Enter | Right-click Start → Command Prompt (or Windows Terminal) | Start Menu → Windows System → Command Prompt. For elevated (Administrator) CMD: right-click Command Prompt → Run as administrator. The title bar shows 'Administrator: Command Prompt' when elevated.

1.2 Anatomy of a CMD Command

```
:: Command structure in Windows CMD:
command [switches/flags] [arguments]

:: Example breakdown:
dir /a /s task_brief.txt
:: dir = command | /a = show all (hidden) | /s = search subfolders |
task_brief.txt = argument

:: Key differences from Linux:
:: Windows uses forward-slash /flag (not dash -flag like Linux)
:: Windows uses backslash \ in paths (Linux uses /)
:: Windows is NOT case-sensitive in commands or filenames
:: C:\Users\Administrator (Windows) vs /home/ubuntu (Linux)
```

CMD Prompt Format

C:\Users\Administrator> — the prompt shows: current drive (C:), full path of current directory (\Users\Administrator), and the > cursor. This is equivalent to ubuntu@tryhackme:~\$ in Linux. There is no root/user distinction shown in the prompt — but running as Administrator means you have elevated privileges.

2. Navigation Commands

Navigating the Windows file system from CMD uses three essential commands: `cd` (to see/change directory), `dir` (to list contents), and a handful of powerful flags that unlock hidden files and recursive search. Together these answer: Where am I? What is here? Where is that file?



cd — Change Directory — show where you are OR move to another folder

Syntax: `cd [path]`

Common flags:

- `cd` with no argument — prints the current directory path (equivalent to `pwd` on Linux)
- `cd foldername` — move into a subfolder
- `cd ..` — go up one directory level (parent folder)
- `cd \` — go to the root of the current drive (C:\)
- `cd /d D:\folder` — change drive AND directory simultaneously (`-d` flag)
- `cd %USERPROFILE%` — go to current user's home folder using environment variable

Examples:

- `cd` :: shows current path e.g.
`C:\Users\Administrator`
- `cd Documents` :: move into Documents
- `cd ..` :: move up to `C:\Users\Administrator`
- `cd C:\Windows\System32` :: navigate using absolute path
- `cd /d D:\Projects` :: change to D: drive and Projects folder

Security use: `cd` to `C:\Windows\System32` reveals core OS files — a favourite location for malware masquerading as system tools. After gaining access to a Windows machine, checking the current directory with bare `cd` confirms exactly where you are before taking further actions.



dir — Directory — list the contents of the current or specified folder

Syntax: `dir [path] [flags]`

Common flags:

- `/a` Show ALL files including hidden and system files (equivalent to `ls -a` on Linux)
- `/s` Search subfolders recursively — also finds files by name across all subdirectories
- `/b` Bare format — filenames only, no size/date info (useful for scripting)
- `/o` Sort output: `/o:n` by name, `/o:s` by size, `/o:d` by date
- `/p` Pause output per screen (like `| less` on Linux)
- `/w` Wide format — multiple columns (quick overview of many files)
- `/q` Show file owner information

Examples:

- `dir` :: list current directory
- `dir /a` :: show ALL including hidden files

- `dir /s task_brief.txt` :: search all subfolders for task_brief.txt
- `dir C:\Users /a /s` :: search entire Users folder including hidden
- `dir /b *.txt` :: list only .txt filenames in current folder
- `dir /a:h` :: show ONLY hidden files

Security use: `dir /a` reveals hidden files that a standard `dir` misses — AppData, hidden system folders, and malware that sets the hidden attribute. `dir /s filename` is the Windows equivalent of `find` — essential for locating files when you only know the name. Security investigators use `dir /s /a` to find suspicious files.

3. Reading Files & Folder Operations

Once you can navigate and find files, the next commands let you read their contents, create directories, copy, move, and delete files — the full set of file management operations from the command line.



type — Display the contents of a text file in the terminal

Syntax: `type [filename]`

Common flags:

- No major flags — `type` simply prints the full file to screen
- `type file1.txt file2.txt` — concatenates and prints both files
- `type file.txt | more` — page through long files (space for next page, q to quit)
- `type file.txt > output.txt` — redirect file content to a new file (copy text)

Examples:

- `type task_brief.txt` :: read and print task_brief.txt
- `type C:\full\path\file.txt` :: read file using absolute path
- `type file.txt | more` :: page through long output
- `type *.txt` :: print all .txt files in current directory

Security use: `type` is the Windows equivalent of `cat`. Security investigators use it to read suspicious scripts, configuration files, and log files. `type C:\Windows\System32\drivers\etc\hosts` reveals the hosts file — a common malware persistence target (fake DNS entries). `type credentials.txt` or `type passwords.txt` are common attacker actions after gaining access.

3.1 File and Directory Management Commands

Command	Syntax	What It Does
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<code>mkdir</code>	<code>mkdir [foldername]</code>	Create a new directory. No -p equivalent — create parent dirs separately or use <code>mkdir a\b\c</code> .
<code>rmdir</code>	<code>rmdir [foldername]</code>	Remove an empty directory. <code>/s</code> removes non-empty directories (with contents). <code>/q</code> quiet mode (no confirmation).
<code>del</code>	<code>del [filename]</code>	Delete one or more files. <code>/f</code> forces deletion of read-only files. <code>/s</code> deletes from all subfolders. <code>/q</code> quiet.
<code>copy</code>	<code>copy [src] [dest]</code>	Copy a file to a new location. <code>/y</code> suppresses overwrite confirmation.
<code>xcopy</code>	<code>xcopy [src] [dest] /e</code>	Extended copy — copies directories and subdirectories. <code>/e</code> includes empty subdirectories. <code>/s</code> skips empty dirs.
<code>move</code>	<code>move [src] [dest]</code>	Move a file or rename it. Works across drives (unlike Linux <code>mv</code> within the same filesystem).
<code>ren</code>	<code>ren [old] [new]</code>	Rename a file or directory without moving it.
<code>echo</code>	<code>echo [text]</code>	Print text to screen. <code>echo text > file.txt</code> writes to file. <code>>></code> appends. <code>echo on/off</code> toggles command echoing.
<code>cls</code>	<code>cls</code>	Clear the screen — equivalent of Linux <code>clear</code> . Cleans up a cluttered terminal.
<code>more</code>	<code>more [file]</code>	Page through long output. Space for next page, Enter for next line, q to quit.

```
:: Create a directory
mkdir C:\Temp\Investigation

:: Copy a file
copy task_brief.txt C:\Temp\backup_task.txt

:: Move/rename a file
move old_name.txt new_name.txt

:: Write text to a file
echo Hello Windows > hello.txt
echo Second line >> hello.txt

:: DANGER — removes directory and all contents without confirmation:
rmdir /s /q C:\Temp\TestFolder

:: Delete all .tmp files in current folder
del /f /q *.tmp
```

4. System Information Commands

After finding and reading task_brief.txt, your next assignment is to gather a system report. These system information commands are the first things IT professionals and security analysts run when they access an unfamiliar Windows machine.

Command	What It Shows	Key Details
<code>whoami</code>	Current username	Prints username of the currently logged-in account. Format: domain\username or computername\username. Essential first check.
<code>hostname</code>	Computer name	Prints the NetBIOS name of the machine. In workplaces, hostnames identify machines on the network. Lab answer: thhmlab
<code>systeminfo</code>	Full system details	Comprehensive output: OS Name, OS Version, System Type (32/64-bit), RAM, hot fixes installed, network adapters. Lab answer for version: 10.0.17763 N/A Build 17763
<code>ver</code>	Windows version (quick)	One-line version output — faster than systeminfo for just the OS version.
<code>ipconfig</code>	Network configuration	Shows IPv4 Address, Subnet Mask, Default Gateway for each adapter. /all flag shows MAC address, DHCP status, DNS servers.
<code>set</code>	Environment variables	Lists all environment variables. Useful: %USERNAME%, %COMPUTERNAME%, %SYSTEMROOT%, %TEMP%, %PATH%.
<code>date /t</code>	Current date	Prints today's date. date without /t lets you change the date.
<code>time /t</code>	Current time	Prints current time. time without /t lets you change the time.
<code>wmic os get caption</code>	OS name (clean)	WMI query for just the OS name — cleaner output than systeminfo for scripting.

```
:: Who am I?
C:\Users\Administrator> whoami
thhm\administrator

:: What machine is this?
C:\Users\Administrator> hostname
thhmlab

:: Full system information (key fields shown):
C:\Users\Administrator> systeminfo
```

```
OS Name:             Microsoft Windows Server 2019 Datacenter
OS Version:          10.0.17763 N/A Build 17763
System Type:          x64-based PC
Total Physical Memory: 4,096 MB

:: Quick OS version only:
C:\Users\Administrator> ver
Microsoft Windows [Version 10.0.17763.7159]
```

5. Network Commands

Windows CMD includes a powerful set of network diagnostic commands. These are used daily by network administrators, incident responders, and penetration testers to understand how a machine is connected and what network activity is occurring.



ipconfig — Show IP address and network adapter configuration

Syntax: `ipconfig [flags]`

Common flags:

- `ipconfig` — basic: IP address, subnet mask, default gateway for each adapter
- `ipconfig /all` — full details: MAC address, DHCP server, DNS servers, lease dates
- `ipconfig /release` — release the current DHCP IP address
- `ipconfig /renew` — request a new DHCP IP address
- `ipconfig /flushdns` — flush the DNS resolver cache (fixes DNS issues)

Examples:

- `ipconfig` :: basic network config
- `ipconfig /all` :: full details including MAC and DNS
- `ipconfig /flushdns` :: clear DNS cache

Security use: `ipconfig /all` reveals the MAC address, DHCP server, and all DNS servers — useful for network mapping and identifying rogue DHCP servers. Attackers run `ipconfig` to understand the network topology after compromising a machine. The Default Gateway reveals the router's IP address.



netstat — Show active network connections and listening ports

Syntax: `netstat [flags]`

Common flags:

- `-a` Show ALL connections and listening ports (not just active)
- `-n` Show addresses as numbers (no DNS lookup — faster)
- `-o` Show the Process ID (PID) responsible for each connection

- `-b` Show the executable name that created each connection (requires Admin)
- `-ano` The most useful combination: all connections, numeric, with PIDs

Examples:

- `netstat -ano` :: all connections with PIDs (most useful)
- `netstat -a` :: all connections and listening ports
- `netstat -b` :: show process names (requires Admin)
- `netstat -ano | find "ESTABLISHED"` :: filter for active connections
- `netstat -ano | find ":80"` :: filter for port 80 activity

Security use: `netstat -ano` is one of the most powerful first commands on a compromised machine. It reveals every active outbound connection (malware C2 channels), every listening port (attack surface), and the PIDs responsible. Cross-reference suspicious PIDs with `tasklist` to identify malicious processes. ESTABLISHED connections to unknown IPs are primary indicators of compromise.



ping — Test network connectivity — ICMP echo request/reply

Syntax: `ping [hostname or IP]`

Common flags:

- `-n count` — number of pings to send (default 4)
- `-t` — ping continuously until Ctrl+C (useful for monitoring)
- `-l size` — packet size to send
- `-a` — resolve IP address to hostname (reverse DNS)

Examples:

- `ping google.com` :: test internet connectivity by name
- `ping 8.8.8.8` :: test connectivity by IP (avoids DNS)
- `ping -n 10 192.168.1.1` :: send 10 pings to the default gateway
- `ping -t 10.10.10.1` :: continuous ping until Ctrl+C

Security use: `ping` confirms whether a remote host is alive. Penetration testers ping target IPs to verify they are reachable. Defenders use `ping -t` to monitor connectivity. Note: many modern servers block ICMP — a non-responsive ping does NOT necessarily mean the host is down.

5.1 Additional Network Commands

Command	Purpose	Key Usage
<code>tracert</code>	Trace route to a destination	<code>tracert google.com</code> — shows every hop between you and the target, latency at each. Windows equivalent of <code>traceroute</code> on Linux.
<code>nslookup</code>	DNS query tool	<code>nslookup google.com</code> — queries DNS for the IP of a domain. <code>nslookup IP</code> — reverse lookup. Useful for DNS investigation.
<code>arp -a</code>	ARP cache / local hosts	Lists IP-to-MAC address mappings cached from recent communications on the local network.

<code>route print</code>	Windows routing table	Shows all routes the OS uses to direct network traffic — equivalent of <code>netstat -r</code> or <code>ip route</code> on Linux.
<code>net use</code>	Network shares	Lists connected network drives. <code>net use Z: \\server\share</code> connects a network drive.
<code>getmac</code>	MAC address of adapters	<code>getmac /v</code> — verbose output showing adapter name, connection status, and MAC address.

6. Process Management Commands

Knowing which processes are running on a Windows machine — and being able to terminate them — is critical for both system administration and incident response. These commands are the CMD equivalents of Task Manager.

Command	Purpose	Key Flags & Usage
<code>tasklist</code>	List all running processes	Shows Image Name, PID, Session, CPU, Memory. <code>/v</code> verbose. <code>/svc</code> shows services. <code>/fi</code> filter (e.g. <code>tasklist /fi "imagename eq notepad.exe"</code>).
<code>taskkill</code>	Terminate a process	<code>/pid [id]</code> — kill by PID. <code>/im [name]</code> — kill by name. <code>/f</code> — force (kills immediately even if unresponsive). e.g. <code>taskkill /im malware.exe /f</code>
<code>sc</code>	Service Control — manage Windows services	<code>sc query</code> — list services. <code>sc start/stop [name]</code> . <code>sc qc [name]</code> — show service config including binary path.
<code>net start</code>	List running Windows services	<code>net start</code> — shows all running services. <code>net stop [name]</code> stops a service. <code>net start [name]</code> starts one.
<code>wmic process</code>	WMI process query	<code>wmic process list brief</code> — lists processes with PID and path. <code>wmic process where name='malware.exe' delete</code> — terminates.

```

:: List all running processes
C:\> tasklist
Image Name          PID      Session  Mem Usage
=====
System Idle Process    0      Services      8 K
System                4      Services     148 K
svchost.exe           752     Services    15,000 K

:: Filter — find a specific process by name
C:\> tasklist /fi "imagename eq notepad.exe"

```

```
:: Kill a process by PID
C:\> taskkill /pid 1234 /f

:: Kill by name (force)
C:\> taskkill /im suspicious.exe /f
```

7. Useful Productivity Commands

Command	Purpose	Details
<code>help</code>	Get help on commands	<code>help</code> — lists all available CMD commands. <code>help dir</code> — detailed help for <code>dir</code> . <code>command /?</code> also shows help (e.g. <code>dir /?</code>)
<code>doskey</code> <code>/history</code>	View command history	Prints all commands run in the current session. Up/Down arrow keys cycle through history inline.
<code>find</code>	Search text in files	<code>find "searchterm" file.txt</code> — search for text in a file. <code>/i</code> case-insensitive. <code>/n</code> show line numbers. <code>/c</code> count occurrences.
<code>findstr</code>	Advanced text search	<code>findstr "pattern" file.txt</code> — supports regular expressions. <code>/s</code> recursive. <code>/i</code> case-insensitive. <code>/n</code> show line numbers.
<code>sort</code>	Sort file output	<code>sort file.txt</code> — sort alphabetically. <code>type file.txt sort</code> — pipe output to sort.
<code>tree</code>	Directory tree view	<code>tree</code> — shows folder structure as a visual tree. <code>tree /f</code> includes filenames in the tree.
<code>attrib</code>	View/change file attributes	<code>attrib file.txt</code> — shows attributes (H=hidden, R=read-only, S=system, A=archive). <code>attrib -h file</code> to unhide.
<code>fc</code>	File compare	<code>fc file1.txt file2.txt</code> — compares two text files and shows differences.
<code>clip</code>	Copy output to clipboard	<code>dir clip</code> — copies <code>dir</code> output to clipboard. <code>type file.txt clip</code> — copies file contents.
<code>exit</code>	Close CMD window	Closes the current CMD session. <code>exit /b</code> in batch scripts exits just the script without closing CMD.

7.1 Piping and Redirection in CMD

Like Linux, Windows CMD supports piping and output redirection — though with slightly different syntax in some cases:

Operator	What It Does & Example
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	Pipe — sends output of left command to right command. <code>tasklist find "chrome"</code> — filter process list.
>	Redirect stdout to file (overwrite). <code>systeminfo > report.txt</code> — save systeminfo output.
>>	Redirect stdout to file (append). <code>echo new line >> existing.txt</code> — add to existing file.
<	Redirect stdin from file. <code>sort < unsorted.txt</code> — sort using file as input.
2>	Redirect error output. <code>dir /s *.txt 2> errors.txt</code> — capture errors separately.
<code>find</code>	Filter piped output for a string. <code>netstat -ano find ":443"</code> — show HTTPS connections only.

8. Linux vs Windows CLI — Side-by-Side Cheat Sheet

Since this room directly follows Linux CLI Basics, here is a complete side-by-side mapping of Linux commands to their Windows CMD equivalents. Both serve the same purpose — the syntax just differs.

Action	Linux (bash)	Windows (CMD)
Show current directory	<code>pwd</code>	<code>cd</code> (with no arguments prints current path)
List directory contents	<code>ls</code>	<code>dir</code>
List including hidden files	<code>ls -la</code>	<code>dir /a</code>
List with details	<code>ls -l</code>	<code>dir</code> (default includes details)
Change directory	<code>cd foldername</code>	<code>cd foldername</code>
Go up one level	<code>cd ..</code>	<code>cd ..</code>
Go to home directory	<code>cd ~</code>	<code>cd %USERPROFILE%</code>
Go to root	<code>cd /</code>	<code>cd \</code>
Find a file by name	<code>find / -name filename</code>	<code>dir /s filename</code> (from starting dir)
Read a file	<code>cat file.txt</code>	<code>type file.txt</code>
Page through file output	<code>less file.txt</code>	<code>more file.txt</code> or <code>type file.txt more</code>
Create a directory	<code>mkdir foldername</code>	<code>mkdir foldername</code>
Delete a file	<code>rm file.txt</code>	<code>del file.txt</code>

Delete directory (recursive)	<code>rm -rf directory</code>	<code>rmdir /s /q directory</code>
Copy a file	<code>cp source dest</code>	<code>copy source dest</code>
Move/rename a file	<code>mv source dest</code>	<code>move source dest</code> or <code>ren oldname newname</code>
Print text	<code>echo text</code>	<code>echo text</code>
Write text to file	<code>echo text > file.txt</code>	<code>echo text > file.txt</code> (same!)
Who am I?	<code>whoami</code>	<code>whoami</code> (same!)
Machine hostname	<code>hostname</code>	<code>hostname</code> (same!)
System/OS info	<code>uname -a</code>	<code>systeminfo</code> or <code>ver</code>
Disk space	<code>df -h</code>	<code>wmic logicaldisk get size,freespace,caption</code>
Running processes	<code>ps aux</code>	<code>tasklist</code>
Kill a process	<code>kill -9 PID</code>	<code>taskkill /pid PID /f</code>
Network config	<code>ip addr</code> or <code>ifconfig</code>	<code>ipconfig</code> or <code>ipconfig /all</code>
Active connections	<code>netstat -tulpn</code>	<code>netstat -ano</code>
Search text in file	<code>grep 'pattern' file</code>	<code>find "pattern" file</code> or <code>findstr</code>
Clear the screen	<code>clear</code>	<code>cls</code>
Command history	<code>history</code>	<code>doskey /history</code> or F7 key
Get help on a command	<code>man command</code>	<code>command /?</code> or <code>help command</code>

9. The Lab Mission — Complete Walkthrough

9.1 Part 1 — Find task_brief.txt

Your boss has hidden task_brief.txt somewhere in the user folder. You only know the filename. Here is the complete step-by-step walkthrough:

Step	Command	What You Learn
1	<code>cd</code>	Check current location — you start in C:\Users\Administrator.
2	<code>dir</code>	List the home directory contents — see 16 directories including Documents, Downloads, etc.

3	<code>dir /a</code>	Show ALL files including hidden — reveals 12+ additional hidden folders not shown by default <code>dir</code> .
4	<code>dir /s task_brief.txt</code>	Search all subfolders from current location for <code>task_brief.txt</code> — Windows prints the full path when found.
5	<code>cd [full path shown by dir /s]</code>	Navigate to the directory containing <code>task_brief.txt</code> using the full path returned by step 4.
6	<code>dir</code>	Confirm <code>task_brief.txt</code> is present in the current directory.
7	<code>type task_brief.txt</code>	Read the file contents — reveals the message TASK-BRIEF-FOUND and your next assignment.

```
:: Step-by-step session (lab answers shown):
```

```
C:\Users\Administrator> cd
C:\Users\Administrator

C:\Users\Administrator> dir /s task_brief.txt
Directory of
C:\Users\Administrator\Documents\Notes\research_yn6\exports_imv\screenshots\notes_wi6

30/01/2026  10:14                84 task_brief.txt

C:\> cd
C:\Users\Administrator\Documents\Notes\research_yn6\exports_imv\screenshots\notes_wi6

C:\...\notes_wi6> type task_brief.txt
TASK-BRIEF-FOUND
Continue to the next task to gather system information.
```

9.2 Part 2 — Gather System Information

The `task_brief.txt` tells you to gather basic system information. These four commands answer the key questions:

Required Info	Command	Lab Answer
Who you are logged in as	<code>whoami</code>	<code>thhm\administrator</code> (or similar)
Name of this computer	<code>hostname</code>	<code>thhmlab</code>
Windows version	<code>systeminfo</code>	<code>10.0.17763 N/A Build 17763</code>
Network configuration	<code>ipconfig</code>	Shows IPv4 address, subnet mask, default gateway

10. Room Answers — Quick Reference

Question	Answer
What is the full path of the task_brief.txt found on the system?	<code>C:\Users\Administrator\Documents\Notes\research_yn6\exports_imv\screenshots\notes_wi6</code>
What message and flag are written inside task_brief.txt?	<code>TASK-BRIEF-FOUND</code>
What is the computer name shown by hostname?	<code>thhmlab</code>
What Windows version is listed in the systeminfo output?	<code>10.0.17763 N/A Build 17763</code>
Which command lists directory contents including hidden files?	<code>dir /a</code>
Which flag makes dir search all subfolders recursively?	<code>/s</code>
Which command reads/displays the contents of a text file?	<code>type</code>
Which command shows current user account name?	<code>whoami</code>

Which command shows network configuration (IP address)?	<code>ipconfig</code>
Which command shows all active connections with PIDs?	<code>netstat -ano</code>

11. Security Implications

CMD Is the Attacker's Post-Exploitation Console

After gaining a foothold on a Windows machine — via phishing, exploit, or credential theft — attackers immediately open CMD or PowerShell. They run `whoami` to confirm their privilege level, `systeminfo` to fingerprint the OS for exploit selection, `ipconfig` to map the network, and `dir /s /a` to hunt for sensitive files. Knowing these commands from both sides is essential.

`dir /a` Reveals What Windows Hides

Windows hides many directories by default: `AppData` (browser credentials, tokens, application data), `$Recycle.Bin` (recently deleted files), and `NTUSER.DAT` (user profile/registry). Malware commonly places files in hidden directories or sets its own files as hidden using `attrib +h`. Security investigators use `dir /a` as standard practice to see everything, not just what Windows shows by default.

`netstat -ano` — The First Incident Response Command

`netstat -ano` is often the first command run when investigating a suspected compromised Windows machine. Any `ESTABLISHED` connection to an unfamiliar external IP is a potential indicator of compromise. Cross-reference the PID from `netstat` with `tasklist` to identify the responsible process. Unusual listening ports (`LISTENING` on `0.0.0.0`) indicate a backdoor or reverse shell listener.

`tasklist` Reveals Malicious Processes

Malware often mimics legitimate process names: `svchost.exe` (legitimate) vs. `scvhost.exe` or `svchost32.exe` (malicious). Legitimate `svchost.exe` lives only in `C:\Windows\System32` — any `svchost.exe` running from another path is suspicious. Use `tasklist /v` to see full paths and correlate with `netstat -ano` PIDs for network-connected processes.

`systeminfo` Reveals Patch Status

The systeminfo output includes installed hotfixes (patches). A machine with few or outdated hotfixes is vulnerable to known exploits — attackers use this to select appropriate privilege escalation exploits. The Hotfix(es) section in systeminfo shows the installed KB numbers — cross-reference these with vulnerability databases like CVEDetails.

The hosts File — A Common Persistence Target

type C:\Windows\System32\drivers\etc\hosts reveals the hosts file — which maps hostnames to IP addresses before DNS is consulted. Malware modifies the hosts file to redirect security update servers to invalid IPs (blocking antivirus updates) or redirect banking sites to fake IPs (pharming). Checking the hosts file should be part of any Windows security investigation.

12. Key Takeaways & Cheat Sheet

12.1 Key Takeaways

- CMD (Command Prompt) is the Windows text-based interface — uses /flags not -flags like Linux, backslash \ not forward slash /.
- cmd bare (no arguments) shows current directory. cd foldername moves into it. cd .. goes up one level.
- dir lists contents. dir /a shows hidden files. dir /s filename searches all subfolders.
- type reads a file (equivalent of Linux cat). type file.txt | more pages through long files.
- dir /s task_brief.txt found the file at:
C:\Users\Administrator\Documents\Notes\research_yn6\exports_imv\screenshots\notes_wi6
- task_brief.txt message: TASK-BRIEF-FOUND
- hostname returns machine name: thhmlab
- systeminfo returns OS version: 10.0.17763 N/A Build 17763
- whoami confirms current user. ipconfig shows network configuration. netstat -ano shows active connections.
- tasklist shows running processes. taskkill /pid PID /f terminates a process.
- Windows pipe (|) and redirect (> >>) work the same as Linux — use find (not grep) to filter piped output.

12.2 Essential CMD Cheat Sheet

Command	What It Does
cd	Show current path (no arg) OR change directory. cd .. = up one level. cd \ = root.
dir	List directory. /a = show hidden. /s = search subfolders. /b = names only.
type [file]	Print file contents to screen. type file.txt more for paging.

<code>mkdir [name]</code>	Create a new directory.
<code>del [file]</code>	Delete a file. /f force. /s subfolders. /q quiet.
<code>copy [src] [dest]</code>	Copy a file.
<code>move [src] [dest]</code>	Move or rename a file.
<code>echo [text]</code>	Print text. > write to file. >> append.
<code>cls</code>	Clear the screen.
<code>whoami</code>	Show current username.
<code>hostname</code>	Show computer name.
<code>systeminfo</code>	Full OS and hardware details including patch level.
<code>ver</code>	Quick Windows version number.
<code>ipconfig</code>	Network config. /all for full details including MAC and DNS.
<code>netstat -ano</code>	All active connections with PIDs — first IR command.
<code>tasklist</code>	All running processes. /fi to filter.
<code>taskkill /pid X /f</code>	Force-terminate a process by PID.
<code>ping [host]</code>	Test connectivity. -n count. -t continuous.
<code>tracert [host]</code>	Trace the network route to a host.
<code>nslookup [domain]</code>	DNS lookup — resolve domain to IP.
<code>find "text" file</code>	Search for text in a file (CMD's grep equivalent).
<code>findstr /i "text"</code>	Advanced text search with regex support.
<code>attrib [file]</code>	Show/change file attributes. -h to unhide.
<code>command /?</code>	Show help for any command (equivalent of man on Linux).

12.3 Next Steps on TryHackMe

- **Immediate:** Continue with the Windows Fundamentals 1, 2 & 3 rooms — deeper Windows CLI, registry, and system tools.
- **Security:** Study the Windows Privilege Escalation room — use `dir /a`, `tasklist`, and `netstat` offensively.
- **Scripting:** Complete the PowerShell Basics room — the more powerful Windows scripting environment.
- **Enterprise:** Take the Active Directory Basics room — Windows in enterprise domain environments.

- **IR:** Work through Incident Response on Windows rooms — using CMD for live forensic investigation.

CMD Is the Common Ground

Whether you are investigating a compromised machine, running a penetration test, or configuring a Windows server, the commands covered in this room are used every time. `dir` /`s` finds evidence. `type` reads logs and scripts. `systeminfo` fingerprints the OS. `netstat -ano` reveals attacker channels. `tasklist` identifies malicious processes. The Windows CMD is the universal tool for all Windows security work.

Source: TryHackMe — Windows CLI Basics Room | tryhackme.com/room/windowsclibasics